Benha University - Faculty of Computers and Artificial Intelligence

Artificial Intelligence Course - 2025/2026

Course Project: Adversarial Search

Team Members:

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1. Project Introduction

For this project, our team selected the game **Connect 4** to implement the concepts of Adversarial Search. The main objective is to design a playable environment where a human player can compete against an AI agent powered by the Minimax algorithm.

2. Game Environment & Mechanics

- **The Board:** The game environment is designed as a 6x7 grid.
- **Input Validation:** The system strictly validates human inputs. It prevents users from playing in full columns and handles invalid data entries (e.g., numbers outside the 0-6 range).
- **Win States:** The environment automatically detects four connected pieces in all possible directions (horizontal, vertical, and diagonal) to declare a winner.

3. The AI Agent & Minimax Algorithm

We implemented the **Minimax Algorithm** to give the computer strategic decision-making capabilities:

- **Game Tree & Depth:** The AI agent anticipates future moves by searching the game tree up to a specific depth (depth = 4) to evaluate the best possible outcomes.
- **Adversarial Strategy:** The algorithm successfully acts as the *maximizing player* to find the winning path, while predicting the human's moves as the *minimizing player* to block any winning threats.
- **Performance Optimization:** Instead of creating multiple copies of the board in memory, the algorithm is optimized to use a "drop and undo" approach, making the AI faster and highly efficient.

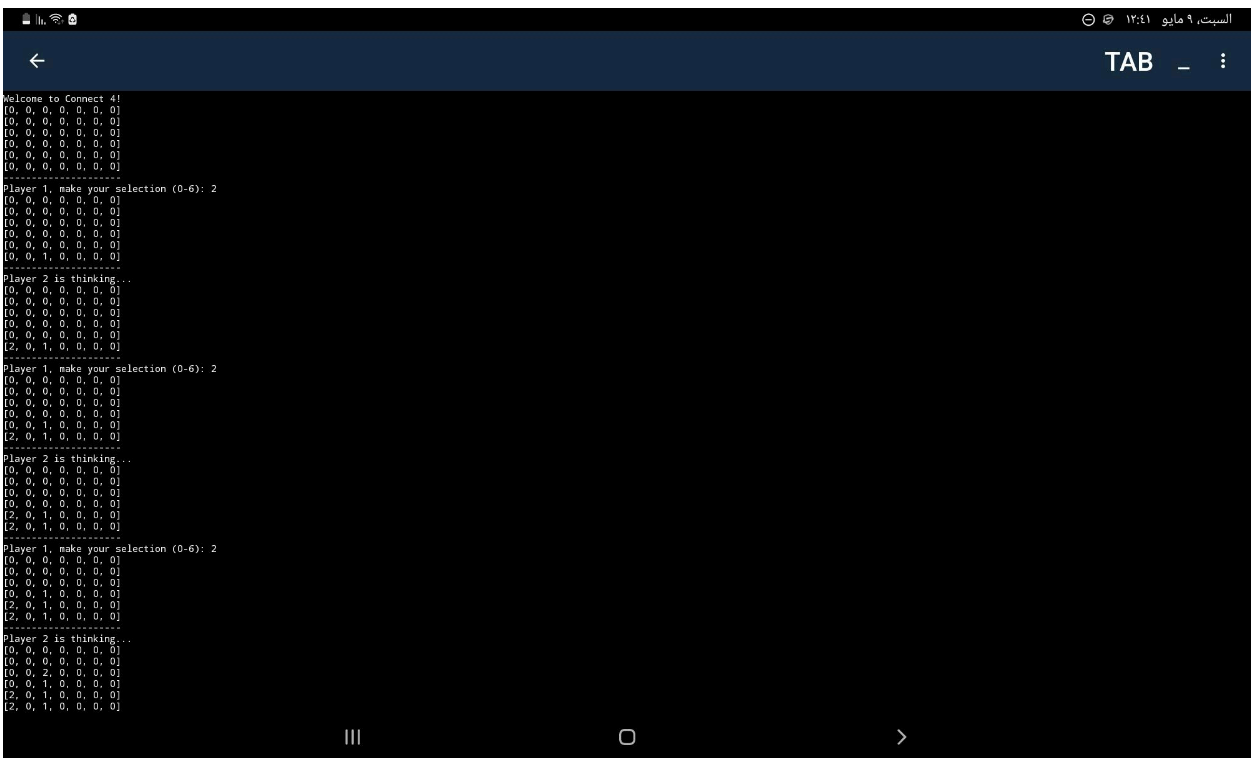
4. Code Structure & Methodology

To ensure a clean, non-redundant codebase, the project was modularized into three main classes, which were divided among the team members:

- **Class Connect4:** Responsible for the grid structure, gravity rules, and win detection.
- **Class MinimaxAI:** Contains the core adversarial logic and the Minimax mathematical evaluation.
- **Class GameManager:** Handles the game loop, turns, and user input validation.

5. Testing & Implementation Results

The project was rigorously tested. The AI agent demonstrated flawless defense mechanisms by successfully blocking the human player's attempts to connect four pieces, while simultaneously building its own offensive strategy to win the game.



TAB

```
[0, 0, 0, 0, 0, 0, 0]
[2, 0, 1, 0, 0, 0, 0]
[2, 0, 1, 0, 0, 0, 0]
---------------------
Player 1, make your selection (0-6): 2
[0, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 0]
[0, 0, 1, 0, 0, 0, 0]
[2, 0, 1, 0, 0, 0, 0]
[2, 0, 1, 0, 0, 0, 0]
---------------------
Player 2 is thinking...
[0, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 0]
[0, 0, 2, 0, 0, 0, 0]
[0, 0, 1, 0, 0, 0, 0]
[2, 0, 1, 0, 0, 0, 0]
[2, 0, 1, 0, 0, 0, 0]
---------------------
Player 1, make your selection (0-6): 2
[0, 0, 0, 0, 0, 0, 0]
[0, 0, 1, 0, 0, 0, 0]
[0, 0, 2, 0, 0, 0, 0]
[0, 0, 1, 0, 0, 0, 0]
[2, 0, 1, 0, 0, 0, 0]
[2, 0, 1, 0, 0, 0, 0]
---------------------
Player 2 is thinking...
[0, 0, 0, 0, 0, 0, 0]
[0, 0, 1, 0, 0, 0, 0]
[0, 0, 2, 0, 0, 0, 0]
[2, 0, 1, 0, 0, 0, 0]
[2, 0, 1, 0, 0, 0, 0]
[2, 0, 1, 0, 0, 0, 0]
---------------------
Player 1, make your selection (0-6): 2
[0, 0, 1, 0, 0, 0, 0]
[0, 0, 1, 0, 0, 0, 0]
[0, 0, 2, 0, 0, 0, 0]
[2, 0, 1, 0, 0, 0, 0]
[2, 0, 1, 0, 0, 0, 0]
[2, 0, 1, 0, 0, 0, 0]
---------------------
Player 2 is thinking...
[0, 0, 1, 0, 0, 0, 0]
[0, 0, 1, 0, 0, 0, 0]
[2, 0, 2, 0, 0, 0, 0]
[2, 0, 1, 0, 0, 0, 0]
[2, 0, 1, 0, 0, 0, 0]
[2, 0, 1, 0, 0, 0, 0]
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Player 2 Wins!

[Program finished]
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